Data Card – TMDB Film Collaborations

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Source TMDB API –> https://api.themoviedb.org/3/discover/movie

Registered scope : Repeated film collaborations in the US between 2020 and 2025, restricted to Drama or Thriller movies, with Director, Producer, Director of Photography and top-billed actors.

Source

The data was collected from the **TMDB API** using:

- `/discover/movie` for movie discovery;
- `/movie/{movie_id}/credits` for cast and crew.

The movie query was restricted to US-origin movies released between 2020-01-01 and 2025-12-31, belonging to Drama (18) or Thriller (53), with at least 100 TMDB votes.

For each movie, the six actors with the lowest billing order were retained, together with crew members whose job was exactly `Director`, `Producer`, or `Director of Photography`.

Collection method & dates

The data was acquired in Python using the `requests` library and TMDB Bearer-token authentication.

The Discover endpoint was paginated with the `page` parameter until all reported pages were retrieved. Credits were then requested once per movie.

Requests were cached locally using the URL and parameters, and retry logic was used for temporary HTTP/network failures.

The acquisition timestamps for movies and credits are stored in `acquisition_log.json`. Raw data was saved as Parquet files.

Scope & size

The final acquired data contains:

- `movies.parquet`: **1208 rows** × **9 columns**;
- `credits.parquet`: **14252 rows** × **7 columns**.

One movie row represents one selected film. One credit row represents one person-role association for a movie.

The selected threshold `vote_count >= 100` was chosen after testing several thresholds. It provides a compromise between excluding very low-engagement titles and keeping enough movies for repeated-collaboration analysis.

Schema

Dataset	Field	Meaning
Movies	id	TMDB movie identifier
	title	Movie title
	original_title	Original title
	release_date	Release date
	genre_ids	List of TMDB genre identifiers
	original_language	Original language code
	vote_count	Number of TMDB votes
	vote_average	Average TMDB rating
	popularity	TMDB popularity score
Credits	movie_id	TMDB movie identifier
	person_id	TMDB person identifier
	person_name	Person name
	role	actor, director, producer or cinematographer
	department	Original TMDB crew department
	job	Original TMDB crew job
	billing_order	Actor billing position; not applicable to crew

Known biases & limitations

The dataset does not represent the entire US film industry.

It excludes movies outside 2020–2025, non-US-origin movies, movies that are neither Drama nor Thriller, and movies with fewer than 100 TMDB votes. The vote threshold therefore favors more visible or engaged-with titles.

Only the top six billed actors are retained, so supporting actors outside these positions are not represented. Crew coverage is also restricted to three exact jobs.

Missing values in the credits table are structural: actors have no `department` or `job`, while crew members have no `billing_order`. These values are kept as not applicable and are not imputed.

The movie variables `vote_count` and `popularity` are strongly right-skewed. Extreme values were retained because they are plausible rather than invalid.

Temporal coverage is complete at the yearly level from 2020 to 2025, but the number of selected movies decreases after 2022 and release counts vary by month.

License / ToS

The data was obtained through the official TMDB API for an academic purpose.

TMDB remains the source of the underlying metadata. API use must comply with TMDB's terms of use and attribution requirements. No TMDB images or other media files were collected.

GDPR note

The credits dataset contains identifiable individuals through `person_id` and `person_name`, corresponding to publicly available professional film credits.

Collection was minimized to information required for collaboration analysis: person identifier, name, role/job, associated movie and billing order where relevant.

No biographies, birth dates, profile images, contact information or social media data were collected.

In our case the remaining identifiers are necessary to distinguish individuals reliably and detect repeated professional collaborations across films.